

An equation of state for predicting the thermodynamic properties and vapour–liquid partitioning of aqueous $\text{Ge}(\text{OH})_4$ in a wide range of water densities



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ARTICLE INFO

Article history:

Received 10 December 2014

Received in revised form 5 February 2015

Accepted 6 February 2015

Available online 7 February 2015

Keywords:

Equation of state
Germanium oxide
Germanic acid
Aqueous solution
Water vapour

ABSTRACT

Although germanic acid is an important compound in many engineering and natural aqueous environments, its exact stoichiometry, structure and stability are insufficiently known. To fill this gap, we combined theoretical quantum chemistry simulations of Ge speciation and structure with a recently developed equation of state for aqueous neutral species, applied to available experimental $\text{GeO}_2(\text{s})$ solubility measurements. Results allow generation of a consistent set of thermodynamic properties of $\text{Ge}(\text{OH})_4(\text{aq})$ over a large T – P –fluid density range (298–900 K, 0.1–300 MPa, and 0.01–1.0 g cm^{−3}). These properties enable quantitative predictions of Ge transport and its fractionation from similar elements (e.g. Si) in aqueous vapour–liquid and supercritical fluid systems.

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1. Introduction

The growing economic demand for critical metals (Sb, Be, Co, Ga, Ge, Li, Mg, Pt group elements, Rare Earth elements, Re, Ta, Nd, Mo, and W) has generated a rapidly increasing interest in geological, chemical and physical communities to almost all aspects of these elements [1]. In particular, knowledge of thermodynamic properties of their solid, aqueous and gaseous forms is required for modelling the concentration and distribution of these elements in mineral deposits, optimizing ore extraction and purification procedures as well as synthesising their compounds in industry. Among these elements, germanium and its oxide GeO_2 are critical components in fibre and infrared optics and polymerization catalysis [2, references therein]. In geochemistry, germanium is an important tracer of continental weathering and riverine flux [3,4], hydrothermal activity [4,5], and organic matter impact [6,7]. This is because in some geological contexts germanium behaves differently from silicon rendering the Ge/Si ratio in waters to be sensitive to the presence of clays and iron hydroxides, organic ligands, reduced sulphur, and temperature [8].

More recently, analytical advances in non-traditional isotope geochemistry have enabled accurate measurements of Ge stable isotopes fractionation upon Ge absorption onto and precipitation with major minerals such as iron hydroxides and clays [9], thus offering additional insights into Ge geochemical cycle.

Quantitative interpretation of Ge transport in natural fluids, its fractionation from silicon, its extraction from ores, and hydrothermal synthesis of its compounds, all require robust data on GeO_2 solubility and Ge aqueous species identity and stability. The available GeO_2 solubility experiments and X-ray absorption spectroscopy studies of Ge-bearing aqueous solutions [4,7,8] unambiguously demonstrate that the uncharged tetra-hydroxide $\text{Ge}(\text{OH})_4$ is by far the dominant species in aqueous solution from acidic to alkaline pH ($1 < \text{pH} < 9$) over a wide temperature, T , (298–723 K) and pressure, P , (0.1–50 MPa) range. The thermodynamic properties of $\text{Ge}(\text{OH})_4(\text{aq})$ derived from $\text{GeO}_2(\text{s})$ solubility measurements using the HKF equation of state [8] are, however, valid only for dense aqueous fluids (pure water density $>0.5 \text{ g cm}^{-3}$) due to intrinsic limitations of the HKF model [10–12]. Other alternative models describing mineral solubilities such as the density model [8,13,14] applied to GeO_2 –water systems may be subjected to large errors beyond the T – P range covered by measurements because of an entirely empirical nature of the model coefficients. In addition, very little is known about Ge hydroxide species identities,

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hydration structure, and stabilities in the low-density vapour phase; this lack hampers quantitative assessment of Ge volatility and vapour–liquid partitioning both in natural hydrothermal–volcanic systems and industrial processes.

To provide a more accurate and quantitative thermodynamic basis for the use of germanium in a variety of natural and industrial applications, we combined theoretical quantum chemistry simulations of Ge speciation and structure with a recently developed equation of state for aqueous neutral species [10,15] based on available experimental solubility measurements. Results allow generation of a consistent set of $\text{Ge}(\text{OH})_4(\text{aq})$ thermodynamic properties over a large T – P –fluid density range and prediction of Ge transport and its fractionation from similar elements (e.g. Si) in aqueous vapour–liquid and supercritical fluid systems.

2. Methods

In this study, we combined several complementary approaches to derive the stoichiometry and structure of Ge hydroxide aqueous and gaseous species, their vapour–liquid partitioning coefficients, and thermodynamic properties.

2.1. Quantum chemistry calculations

We carried out a set of quantum chemical computations of the structure and thermochemical properties of ideal gas Ge hydroxides and their different water clusters: $\text{GeO}(\text{OH})_2$, $\text{Ge}(\text{OH})_4$, $\text{Ge}(\text{OH})_4 \cdot \text{H}_2\text{O}$, and $\text{Ge}(\text{OH})_4 \cdot (\text{H}_2\text{O})_2$. Computations were performed with the Gaussian 09W, Revision C.01 suite of programs [16], using the 6-311G+(d, p) basis set. Electron correlation effects were accounted for by the use of the density functional theory (DFT) with the hybrid functional B3LYP, which has been chosen as the most optimal compromise between precision and computing time. All molecular structures were processed and analysed for imaginary frequencies using the GaussView 5 program [16]. The structure optimization procedure was followed by thermochemical calculations of the species under consideration. Additional computations for $\text{Ge}(\text{OH})_4$ have also been performed with higher level of theory using CBS and G3MP2 methods. The computed Ge–O bond distances are 1.770 and 1.763 Å for CBS–QB3 and G3MP2 respectively, which are close to the 1.772 Å value computed by the DFT method chosen above. The computed values of entropy at 298.15 K are 359.64 and 360.03 $\text{J mol}^{-1} \text{K}^{-1}$ for CBS–QB3 and G3MP2 respectively, which is in excellent agreement with the value of 360.07 $\text{J mol}^{-1} \text{K}^{-1}$ (Table 4) computed by the DFT method. The temperature dependence of thermochemical properties was computed using a standard Gaussian procedure (the Read isotopes option) in the temperature range of 298–1500 K. No corrections for hindered rotation and anharmonicity of thermal vibrations as well no scaling factors for vibration frequencies were applied. The computed values of heat capacities of the species, C_p ,

at different T points were least-square regressed to generate parameters for the C_p temperature dependence. The overall relative error in this regression does not exceed 0.3% in a whole temperature range. The optimized structures of the different species considered are shown in Fig. 1 and the computed thermodynamic properties are reported in Table 1

2.2. Equations describing vapour–liquid partitioning

One of the challenges of any equation of state for aqueous species is to accurately predict the behaviour of solutes near the critical point of water. The thermodynamics of dilute near-critical solutions is governed by the so-called Krichevskii parameter [18]. For example, variations of the vapour–liquid distribution constant at infinite dilution, K_D , at temperatures, T , from 450–500 K up to the critical point of water (647 K) are accurately described by the relation given by Japas and Levett Sengers [19]:

$$RT \ln K_D = 2A_{\text{Kr}} \frac{(\rho_1^*(L) - \rho_{1c})}{\rho_{1c}^2} M_w, \quad (1)$$

where $R = 8.31446 \text{ J mol}^{-1} \text{K}^{-1}$, $K_D = \lim_{x \rightarrow 0} y/x$, and y and x are the mole fractions of the solute in the coexisting liquid and vapour phases of water, respectively, $\rho_{1c} = 0.322 \text{ g cm}^{-3}$ is the critical water density, $\rho_1^*(L)$ is the saturated liquid water density (in g cm^{-3}), the molar mass of water $M_w = 18.0153 \text{ g mol}^{-1}$, and A_{Kr} is the Krichevskii parameter (in MPa). In the equations that follow, the superscripts * and ∞ are used to designate the properties of a pure substance and of a solute at infinite dilution, respectively, according to IUPAC recommendations. The subscripts 1 and 2 designate water and a solute.

In the absence of direct experimental measurements on Ge vapour–liquid partitioning, the values of $\ln K_D$ were estimated from the solubility of GeO_2 in the liquid phase [4] and in the unsaturated vapour phase [20] using different approximations whose results are described in Section 3.2.

2.3. Equation of state

The standard state convention adopted in this work is the one defined by IUPAC [21]. For a gaseous substance it is the hypothetical state of the pure substance in the gaseous state at the standard pressure $P^\circ = 0.1 \text{ MPa}$ and exhibiting an ideal gas behaviour. The standard state for solid phases is a pure substance at 0.1 MPa, the standard state for solutes is an aqueous solution of solute molality $m = 1$, at a given pressure P , exhibiting infinitely diluted solution behaviour.

The detailed derivation of the EoS used in this study is given by Akinfiev and Diamond [10]. Briefly, the chemical potential of an aqueous nonelectrolyte species at infinite dilution at a given temperature T (in K) and pressure P (in MPa) is expressed as

Table 1
Thermodynamic parameters of H_2O and germanium hydroxides computed in this study.

Species	Ideal gas properties			
	$(\text{H}_a)_{298}^\circ$, Hartree mol^{-1}	S_{298}° , $\text{J mol}^{-1} \text{K}^{-1}$	$C_{p,298}$, $\text{J mol}^{-1} \text{K}^{-1}$	$C_p(T)^\text{b}$, $\text{J mol}^{-1} \text{K}^{-1}$
$\text{H}_2\text{O}^\text{c}$	–76.45482	188.64	33.48	$29.68 + 8.45 \times 10^{-3} \times T + 0.84 \times 10^5 \times T^{-2} + 4.21 \times 10^{-6} \times T^2 - 1.74 \times 10^{-9} \times T^3$
$\text{Ge}(\text{OH})_4$	–2380.50134	360.07	121.91	$119.82 + 101.08 \times 10^{-3} \times T - 20.21 \times 10^5 \times T^{-2} - 65.75 \times 10^{-6} \times T^2 + 17.92 \times 10^{-9} \times T^3$
$\text{GeO}(\text{OH})_2$	–2303.98027	310.03	80.38	$75.46 + 78.01 \times 10^{-3} \times T - 12.55 \times 10^5 \times T^{-2} - 53.66 \times 10^{-6} \times T^2 + 14.43 \times 10^{-9} \times T^3$
$\text{Ge}(\text{OH})_4(\text{H}_2\text{O})$	–2456.95344	417.90	158.97	$152.30 + 134.91 \times 10^{-3} \times T - 24.36 \times 10^5 \times T^{-2} - 78.43 \times 10^{-6} \times T^2 + 20.17 \times 10^{-9} \times T^3$
$\text{Ge}(\text{OH})_4(\text{H}_2\text{O})_2$	–2533.40577	470.97	195.42	$184.23 + 170.07 \times 10^{-3} \times T - 28.63 \times 10^5 \times T^{-2} - 91.77 \times 10^{-6} \times T^2 + 22.53 \times 10^{-9} \times T^3$

^a Sum of electronic and thermal free energies computed in Gaussian.

^b Temperature range for $C_p(T)$ equation 298–1500 K.

^c For comparison, CODATA values [17] are $S_{298}^\circ = 188.84 \pm 0.01 \text{ J mol}^{-1} \text{K}^{-1}$, $C_{p,298} = 33.43 \text{ J mol}^{-1} \text{K}^{-1}$.

Table 2Experimental data on solid GeO₂ solubility used in the present study.

Solid phase	Temperature range/K	Pressure range/MPa	Density of the solvent, g cm ⁻³	N ^a	Reference
GeO ₂ (tetr)	623–773	6.5–300	0.02–0.85	47	Glemser et al. [20]
GeO ₂ (tetr)	298	0.1	1.0	1	Müller [27]
GeO ₂ (tetr)	298–573	P _{sat.}	0.7–1.0	7	Kossova et al. [28]
GeO ₂ (tetr)	298–623	P _{sat.}	0.57–1.0	8	Pokrovski and Schott [4]
GeO ₂ (tetr)	573–773	40	0.18–0.76	5	Pokrovski et al. [8]
GeO ₂ (hex)	298	0.1	1.0	1	Müller [27]
GeO ₂ (hex)	298	0.1	1.0	1	Pugh [29]
GeO ₂ (hex)	284–314	0.1	1.0	5	Schwartz and Huf [30]
GeO ₂ (hex)	298	0.1	1.0	1	Laubergayer and Morton [31]
GeO ₂ (hex)	313–373	0.1	1.0	4	Evdokimov and Kogan [32]
GeO ₂ (hex)	298	0.1	1.0	1	Gayer and Zajicek [33]
GeO ₂ (hex)	273–373	0.1	1.0	5	Vehov et al. [34]

^a Number of averaged *T*–*P* data points reported in the original publications.**Table 3**Thermodynamic properties of the solid phases GeO₂(tetr) and GeO₂(hex) adopted in this study from Pokrovski and Schott [4].

Phase	$\Delta_f G_{298}^\circ$, kJ mol ⁻¹	$\Delta_f H_{298}^\circ$, kJ mol ⁻¹	S_{298}° , J mol ⁻¹ K ⁻¹	V_{298}° , cm ³ mol ⁻¹	C_p^a , J mol ⁻¹ K ⁻¹
GeO ₂ (tetr)	–521.36 ± 2.93	–580.32 ± 2.93	39.71 ± 0.17	17.42	68.86 + 9.11 × 10 ⁻³ × <i>T</i> – 1.90 × 10 ⁶ × <i>T</i> ⁻²
GeO ₂ (hex)	–500.82 ± 1.25	–554.80 ± 1.25	55.27 ± 0.42	24.44	61.18 + 13.79 × 10 ⁻³ × <i>T</i> – 1.17 × 10 ⁶ × <i>T</i> ⁻²

^a Temperature range for the *C_p*(*T*) equation is 298–1300 K.

$$g_{2,aq}^\infty(P, T) = g_{2,g}^\circ(T) - RT \ln N_w + (1 - \xi) RT \ln \frac{f_1^*}{P^\infty} + RT \xi \ln \left(\frac{RT}{P^\infty M_w} \rho_1^* \right) + RT \rho_1^* \left[a + b \left(\frac{10^3}{T} \right)^{0.5} \right]. \quad (2)$$

Here the pure water density ρ_1^* is expressed in g cm⁻³, the pure water fugacity f_1^* – in MPa, $R = 8.31446$ J mol⁻¹ K, the molar mass of water $M_w = 18.0153$ g mol⁻¹, the number of moles of water in 1 kg of water $N_w = 1000/M_w \approx 55.508$, and $P^\infty = 0.1$ MPa is the standard pressure. Symbols $g_{2,aq}^\infty(T, P)$ and $g_{2,g}^\circ(T)$ stand for the chemical potential of a solute in aqueous solution at infinite dilution and in ideal gas state, respectively. The symbols ξ , a , and b denote empirical parameters independent of P and T : ξ is a “scaling” factor proposed by Plyasunov et al. [22], $(1 - \xi)$ is the measure of the relative size of the solute molecule compared to the solvent molecule, and a and b account for the temperature effects of hydration of the solute. Eq. (2) can be modified to express the Henry constant of the solute, $k_H(T, P)$:

$$\ln(k_H) = (1 - \xi) \ln \frac{f_1^*}{P^\infty} + \xi \ln \left(\frac{RT}{P^\infty M_w} \rho_1^* \right) + \rho_1^* \left[a + b \left(\frac{10^3}{T} \right)^{0.5} \right], \quad (3)$$

and its distribution coefficient at infinite dilution, $K_D(T, P_{sat.})$, between the coexisting vapour and liquid phases

$$\ln K_D = -\xi \ln \frac{\rho_{1,vap}^*}{\rho_{1,liq}^*} - \left[a + b \left(\frac{10^3}{T} \right)^{0.5} \right] (\rho_{1,vap}^* - \rho_{1,liq}^*). \quad (4)$$

Derivation of the expressions for k_H , K_D and derivatives of Eq. (2) corresponding to the partial molar volume and heat capacity of the solute are given in [15], together with expressions for the solute fugacity coefficient at infinite dilution in water, φ_2^∞ , which is a property reflecting deviations from ideality of water-solute mixtures [23]. At near-critical conditions, the partial molar properties of a solute at infinite dilution in water are governed by the Krichevskii parameter, A_{Kr} , and the corresponding analytical expressions are given by Levelt Sengers [18].

The use of Eq. (2) for volatile solutes, when the properties of the corresponding gaseous species are well known, allows for an accurate description of their thermodynamic properties over a wide range of temperatures (273–873 K) and pressures (0.1–300 MPa), including the near-critical region of the solvent [10]. This EoS also yields the correct low pressure limit for all thermodynamic properties of the solute; thus, it can also be employed in the low-density region of the solvent. As follows from Eq. (2), to compute $g_{2,aq}^\infty(P, T)$, in addition to $g_{2,g}^\circ(T)$, ξ , a , and b , one needs to know the solvent thermodynamic properties, f_1^* and ρ_1^* , at the specified P – T point. Here we have employed the most reliable

Table 4

Thermodynamic parameters of Ge and Si hydroxides adopted in this study.

Species	Ideal gas properties			EoS parameters		
	$\Delta_f G_{298}^\circ$, kJ mol ⁻¹	S_{298}° , J mol ⁻¹ K ⁻¹	$C_p(T)^{a,b}$, J mol ⁻¹ K ⁻¹	ξ	a , cm ³ g ⁻¹	b , cm ³ K ^{0.5} g ⁻¹
Ge(OH) ₄	–896.6 ± 3.0 ^c	360.07	119.6 + 0.102 × <i>T</i> – 2.021 × 10 ⁶ × <i>T</i> ⁻² – 6.575 × 10 ⁻⁵ × <i>T</i> ² + 1.792 × 10 ⁻⁸ × <i>T</i> ³	–2.38 ± 0.13	–10.3 ± 1.6	7.9 ± 1.6
Si(OH) ₄	–1239.66 ± 1.17 ^d	346.37	96.1 + 0.158 × <i>T</i> – 1.858 × 10 ⁶ × <i>T</i> ⁻² – 1.162 × 10 ⁻⁴ × <i>T</i> ² + 3.315 × 10 ⁻⁸ × <i>T</i> ³	–1.89 ± 0.07	0.928 ± 0.063	–0.941 ± 0.049

^a Quantum chemical values.^b Temperature range for the *C_p*(*T*) equation is 298–1500 K.^c Includes uncertainty contributions both from the linear fit (±0.5 kJ mol⁻¹) and from the Gibbs energy of solid phases of GeO₂ (±2.9 kJ mol⁻¹). Error corresponds to the 2σ confidence using adopted experimental data on GeO₂(cr.) solubility and the mean $\Delta_f G_{298}^\circ$ (GeO₂(cr.)) values from Table 1.^d Includes uncertainty contributions both from the linear fit (±0.6 kJ mol⁻¹) and $\Delta_f G_{298}^\circ$ (Quartz) (±1.0 kJ mol⁻¹).

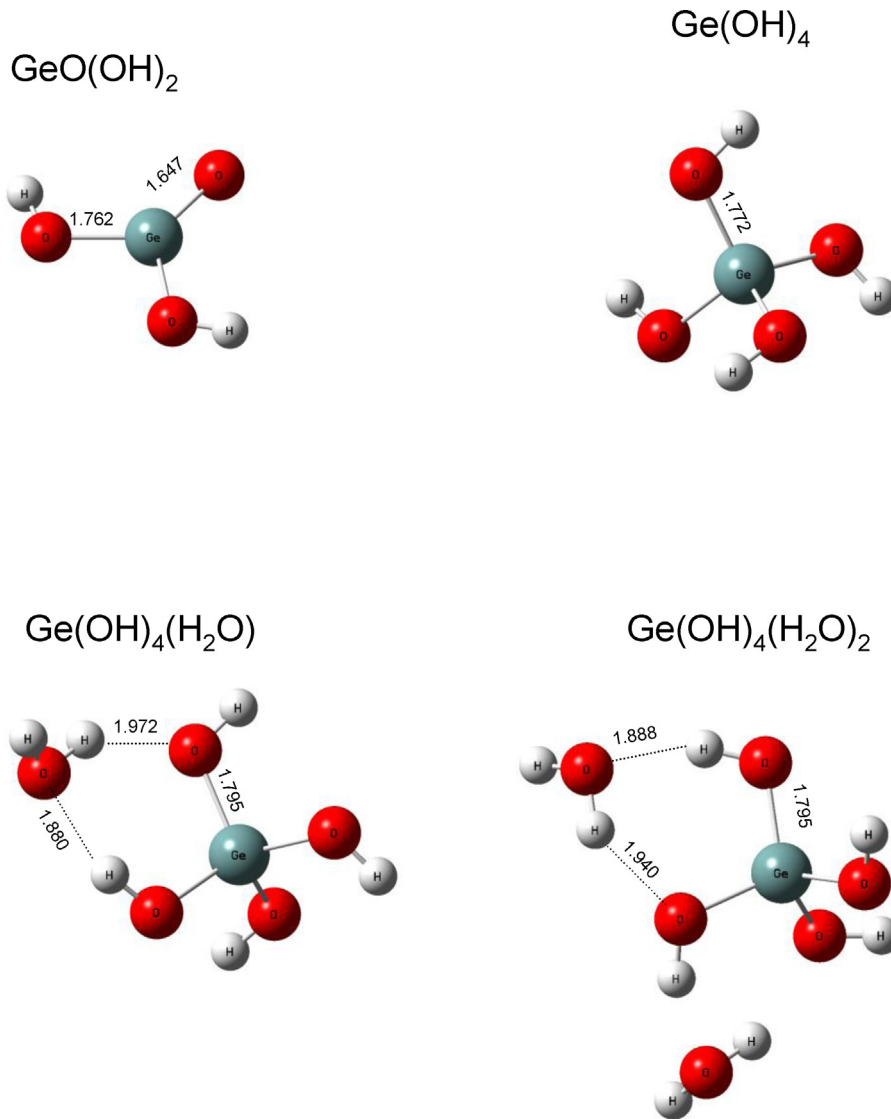


Fig. 1. Optimized geometries for the indicated Ge hydroxide species. Bond distances are in Å.

Wagner and Pruß formulation [24] recommended by IUPAC that is valid from the triple point pressure and temperature to at least 1000 MPa (or the melting curve) and 1273 K. Because of a lack of the thermodynamic properties in the ideal gas state for nonvolatile Ge-bearing species, the quantum chemical computations were employed to determine their entropy, $S_{2,g}^0(298)$, and the temperature dependence of their heat capacity, $C_{p2,g}^0(T)$. They thus allow for expression of, $g_{2,g}^0(T)$ in a “traditional” way

$$g_{2,g}^0(T) = g_{2,g}^0(T_r) - S_{2,g}^0(T_r) \times (T - T_r) + \int_{T_r}^T C_{p2,g}^0(T) dT - T \int_{T_r}^T \frac{C_{p2,g}^0(T)}{T^2} d(\ln T), \quad (5)$$

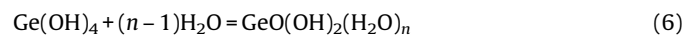
where T_r is the standard temperature of 298.15 K. Combining (2) and (5) results in a linear dependence between $g_{2,aq}^\infty(T, P)$ and $g_{2,g}^0(T)$, allowing the $g_{2,g}^0(T_r)$ and the EoS parameters, ξ , a , and b , to be determined by regression of the available experimental $g_{2,aq}^\infty(T, P)$ data.

3. Results

3.1. Germanium hydroxides structure and stability in the gas phase and aqueous solution

The computed Ge(OH)_4 structure in the ideal gas state (Fig. 1) represents a symmetrical GeO_4 tetrahedron, with Ge–O bond distances of 1.77 ± 0.02 Å, which are identical within errors to those experimentally determined by X-ray absorption spectroscopy (XAS) in aqueous solution of germanic acid in a wide T – P range (1.75 ± 0.01 Å) [4]. The presence of additional water molecules, forming a 2nd shell around Ge(OH)_4 , results in lengthening of the Ge–O bond distances and lowering of the corresponding vibrational frequency. In contrast, partially dehydrated clusters like GeO(OH)_2 have a planar structure with shorter Ge–O bonds.

The computed thermodynamic properties for the different hydrated clusters (Table 1) enable estimation of the relative stability of the species in aqueous solution according to the reaction



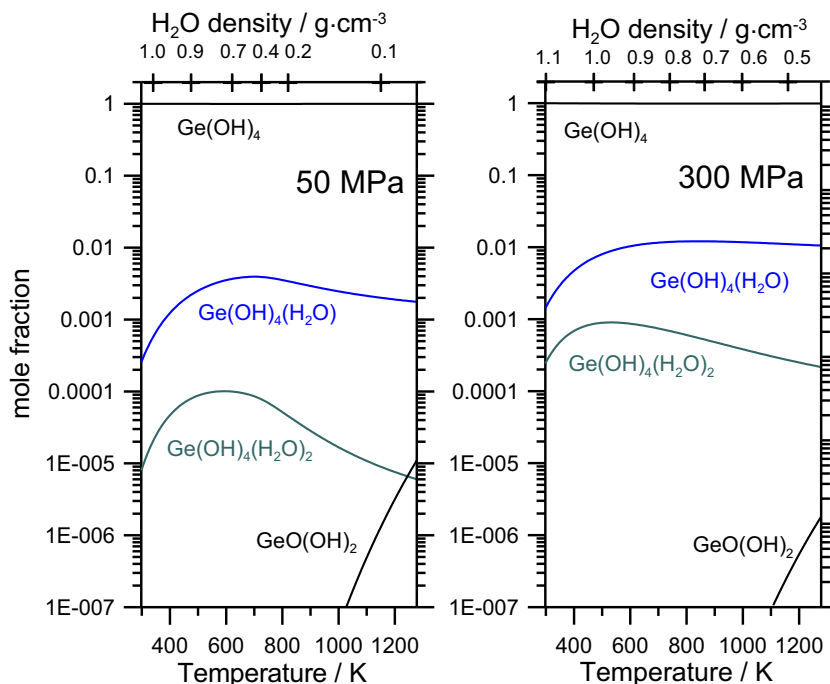


Fig. 2. Distribution of Ge hydroxide species in aqueous fluid as a function of temperature at indicated pressures. Fugacity coefficients of different Ge-bearing species are assumed to be equal to one another (see text).

with $n = 0 - 3$ whose equilibrium constant K_n^o is expressed as

$$\ln K_n^o = \frac{p_{\text{GeO(OH)}_2(\text{H}_2\text{O})_n} \times \phi_{\text{GeO(OH)}_2(\text{H}_2\text{O})_n}}{p_{\text{Ge(OH)}_4} \times \phi_{\text{Ge(OH)}_4} \times (f_1^*)^{n-1}} = \frac{g_{\text{Ge(OH)}_2(\text{H}_2\text{O})_n} - g_{\text{Ge(OH)}_4} - (n-1) \times g_{\text{H}_2\text{O}}}{RT}, \quad (7)$$

where p stands for the species partial pressure, ϕ denotes the fugacity coefficient, f_1^* is the fugacity of H_2O at given pressure P and temperature T , and g stands for the standard molal Gibbs energy of the subscripted species in the ideal gas state at given T . The water fugacity at a given T - P point was computed after the Wagner and Pruß (2002) formulation [24]. For a rough, within an order-of-magnitude, estimation, one can assume ϕ of the different germanium hydrates in an aqueous environment to be equal to one another. The molal fractions of the various hydrates in a gaseous mixture, calculated using Eq. (7), are shown in Fig. 2 as a function of temperature at 50 and 300 MPa. It can be seen that the mole fraction of anhydrous Ge(OH)_4 is very close to 1. Note, that the increase of pressure (or H_2O density) increases only slightly the mole fractions of hydrated Ge(OH)_4 clusters, so that their contribution remains to be less than 1% of the total Ge in the whole temperature and pressure range, and thus will be ignored in the subsequent analysis. Thus, our calculations suggest that Ge(OH)_4 is by far the major Ge-bearing species in aqueous environments.

3.2. Estimation of the Krichvskii parameter of Ge(OH)_4

At low Ge concentrations in solution in equilibrium with a solid phase, the partition coefficient in Eq. (4) may be approximated by

$$K_D \approx \frac{y_s}{x_s}, \quad (8)$$

where y_s and x_s stand for the solubility of GeO_2 in the coexisting water vapour and liquid phases, respectively. The solubility of GeO_2 in liquid water (m) at 623.2 K is $0.0209 \text{ mol kg}^{-1}$ [4]; therefore, $x_s = m/55.508 \approx 3.77 \times 10^{-4}$. In the absence of direct vapour–

liquid partitioning data, the y_s value was determined from Glemser et al.'s data [20] on tetragonal GeO_2 (argutite) solubility in undersaturated steam at 623.2 K and 6.5 MPa ($y_s = 5.86 \times 10^{-7}$), 9.4 MPa ($y_s = 1.26 \times 10^{-6}$), 11.0 MPa ($y_s = 2.20 \times 10^{-6}$) by extrapolating to $P_{\text{sat}} = 16.54 \text{ MPa}$ at 623.2 K using two approaches.

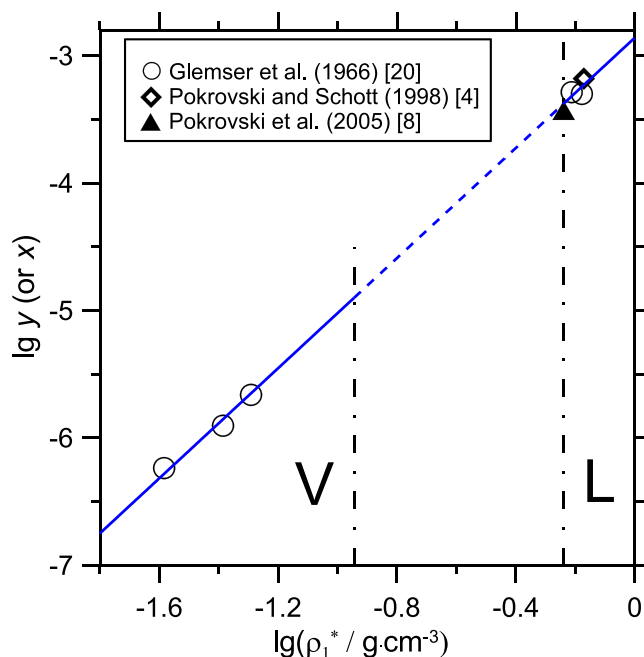


Fig. 3. The estimation of $\lg y_s$ by means of the "density" Eq. (10) at 623.2 K (see text). Symbols x and y stand for the mole fractions of Ge in the liquid (L) and vapour (V) phases of water, respectively: $\circ$, Glemser et al. [20]; $\diamond$, Pokrovski and Schott [4]; $\blacktriangle$, Pokrovski et al. [8]. The vertical dash-dotted lines delimit the densities of the coexisting V and L phases at saturation pressure. The straight solid line indicates the predicted values of $\lg x$ or $\lg y$ according to Eq. (12) (the dashed region of the line indicates the density gap between the vapour and liquid phase and is shown as an eye guide).

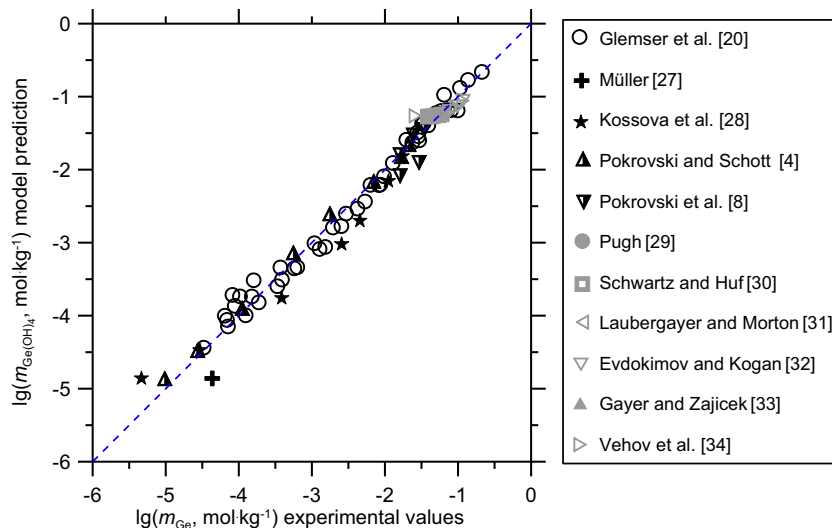


Fig. 4. Comparison between the experimental GeO_2 solubilities in water and the predicted values using the EoS from this study. The whole set of available experimental data covers the range of temperatures of 280–773 K, pressures 0.1–300 MPa, and solvent densities 0.02–1 g cm^{-3} : $\circ$, Glemser et al. [20]; $+$, Müller [27]; $\star$, Kossova et al. [28]; Δ , Pokrovski and Schott [4]; ∇ , Pokrovski et al. [8]; $\bullet$, Pugh [29]; $\square$, Schwartz and Huf [30]; $\triangleleft$, Laubergayer and Morton [31]; $\triangledown$, Evdokimov and Kogan [32]; $\blacktriangle$, Gayer and Zajicek [33]; $\triangleright$, Vehov et al. [34]. The black symbols correspond to the solubility of tetragonal GeO_2 (argutite), while the grey symbols denote the solubility of the hexagonal polymorph. The dashed line corresponds to the ideal agreement between experiment and prediction.

The first approach uses the solubility reaction



whose equilibrium constant is expressed as¹

$$\ln K^\circ = \ln \frac{y_2 \phi_2^\infty P^\circ}{P(\phi_1^*)^2} - \frac{V_s(P - P^\circ)}{RT}, \quad (10)$$

where ϕ_2^∞ is the fugacity coefficient of $\text{Ge}(\text{OH})_4$ at infinite dilution in water; ϕ_1^* is the fugacity coefficient of pure water; y_2 is the mole fraction of $\text{Ge}(\text{OH})_4$ (the mole fraction of water is taken equal to one due to the small solubility of the solid phase); $P^\circ = 0.1$ MPa; $V_s = 17.42 \text{ cm}^3 \text{ mol}^{-1}$ [4] is the molar volume of the tetragonal $\text{GeO}_2(\text{s})$. The fugacity coefficient of $\text{Ge}(\text{OH})_4$ at infinite dilution in water is estimated according to Akinfiev and Plyasunov [15]:

$$\ln \phi_2^\infty \approx 7 \ln \phi_1^*. \quad (11)$$

The three experimental solubility data points of Glemser et al. [20] cited above were used to estimate the value of $\lg K^\circ = -8.26 \pm 0.08$ (2σ), which corresponds to $\lg y_s = -5.24 \pm 0.08$ at 623.2 and 16.54 MPa.

The second approach uses the well-known “density model”, that suggests the following relationship between the solubility of a solid phase and the water density at a given temperature [25]:

$$\lg y_s = A + B \lg \rho_1^*, \quad (12)$$

where A and B are empirical P -independent parameters, and ρ_1^* is the density of pure water (in g cm^{-3}). A similar relation holds in the GeO_2 – H_2O system at 623 K, see Fig. 3, with $A = -2.86 \pm 0.06$ (2σ) and $B = 2.16 \pm 0.07$ (2σ); this corresponds to $\lg y_s = -4.90 \pm 0.12$.

Applying Eqs. (1) and (8) with the y_s values obtained using the two approaches above yields two values of the Krichevskii parameter, A_{Kr} , for aqueous $\text{Ge}(\text{OH})_4$: -247 ± 11 and -201 ± 16 MPa. The agreement is better than 20% of the value. Experimental determinations of the $\text{Ge}(\text{OH})_4$ distribution between the aqueous vapour and liquid phases would be very helpful for a more precise

estimation of the Krichevskii parameter for this species. For comparison, the recommended value of A_{Kr} for $\text{Si}(\text{OH})_4$ is -190 ± 10 MPa [26], which implies that $\text{Si}(\text{OH})_4$ is slightly more volatile than $\text{Ge}(\text{OH})_4$.

3.3. Derivation of the equation-of-state parameters for $\text{Ge}(\text{OH})_4$

The whole set of the available experimental data of GeO_2 solubility (see Table 2) was used to evaluate the parameters of the equation of state $g_{2,g}^\circ(T_r)$, ξ , a , and b . These data are in reasonable agreement with one another and cover a wide range of temperatures (273–773 K), pressures (0.1–300 MPa), and densities of water (0.02–1 g cm^{-3}); they allow calculation of $g_{2,g}^\circ$ of $\text{Ge}(\text{OH})_4(\text{aq})$ according to the reactions



$$g_{2,g}^\circ(\text{Ge}(\text{OH})_4(\text{aq}))(T, P) = g(\text{GeO}_2(\text{s}))(T, P) + 2g(\text{H}_2\text{O})(T, P) - RT \ln m_{\text{Ge}}, \quad (14)$$

where m_{Ge} corresponds to the experimental molality of germanium in aqueous solution, and g is the standard molal Gibbs free energy of the indicated compound. The g values of germanium oxide, $\text{GeO}_2(\text{s})$, (both tetrahedral and hexagonal polymorphs) were generated using the thermodynamic data reported by Pokrovski and Schott [4] (Table 3); those of H_2O were computed according to the Wagner and Pruß [24] formulation.

The regression procedure to obtain the $g_{2,g}^\circ(\text{Ge}(\text{OH})_4)$ value at the standard state conditions (298 K, 0.1 MPa) as well as the parameters ξ , a , and b , is based on Eqs. (2) and (5) and uses quantum chemical value of the entropy, S_{298}° , and heat capacity, $C_p(T)$, from Table 1. The retrieved values for gaseous $\text{Ge}(\text{OH})_4$ are: $g_{2,g}^\circ(T_r) = \Delta_f G_{298}^\circ = -896.6 \pm 0.50 \text{ kJ mol}^{-1}$, and the EoS parameters are $\xi = -2.38 \pm 0.13$, $a = -10.3 \pm 1.6 \text{ cm}^3 \text{ g}^{-1}$, and $b = 7.9 \pm 1.6 \text{ cm}^3 \text{ K}^{0.5} \text{ g}^{-1}$ ($\pm 2\sigma$, Table 4). An independent check of the validity of these coefficients is the resulting value of the Krichevskii parameter of $\text{Ge}(\text{OH})_4(\text{aq})$, $A_{Kr} = -245.2 \pm 13.4$ MPa, which is identical within errors to one of the values estimated above (Section 3.2). It can be seen in Fig. 4 that the whole set of experimental data included in the regression procedure is well

¹ For practical reasons and to facilitate comparisons with the available data, numerical values of equilibrium constants, molality and density coefficients in the following text and figures are expressed in decimal log scale ($\lg$), while the original thermodynamic equations preserve natural logarithm notation ($\ln$).

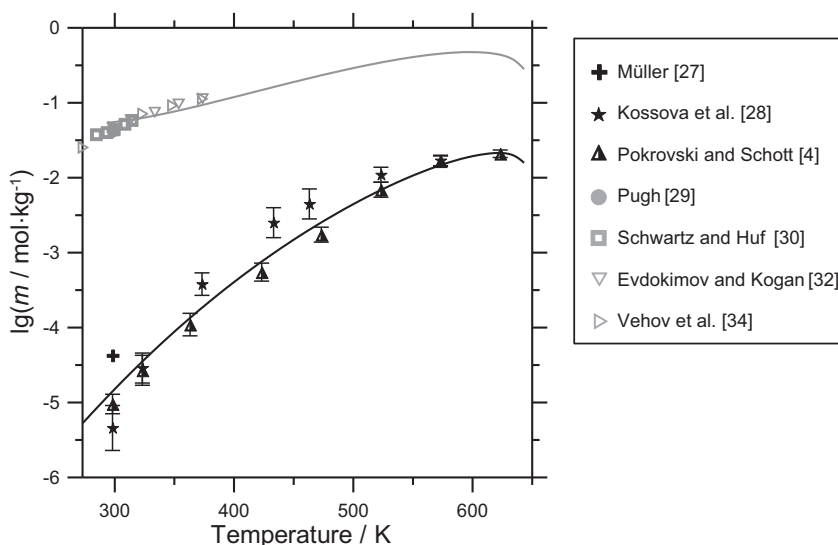


Fig. 5. Solubility of tetrahedral (black symbols) and hexagonal (grey symbols) GeO_2 in liquid water as a function of temperature at saturation water pressure (P_{sat}). Symbols correspond to experimental data: $\blackplus$, Müller [27]; $\blackstar$, Kossova et al. [28]; $\blacktriangle$, Pokrovski and Schott [4]; $\bullet$, Pugh [29]; $\blacksquare$, Schwartz and Huf [30]; ∇ , Evdokimov and Kogan [32]; $\triangleright$, Vehov et al. [34], curves are model predictions.

described by the model. Fig. 5 illustrates the agreement between the experimental data and model predictions as a function of temperature at saturation water pressure, while Fig. 6 – as a function of pressure at 673 and 773 K. These comparisons demonstrate the performance of the proposed model over a wide range of T – P – ρ_1^* conditions.

4. Discussion

The capabilities of our EoS for predicting $\text{Ge}(\text{OH})_4$ thermodynamic properties may be compared with the classical existing equations, the HKF [35] and the density [13,14] models whose coefficients were reported by Pokrovski et al. [8] from $\text{GeO}_2(\text{s})$ solubility measurements. The comparison of the three models with experiment is illustrated in Fig. 7 using the $\text{GeO}_2(\text{s})$ solubility data

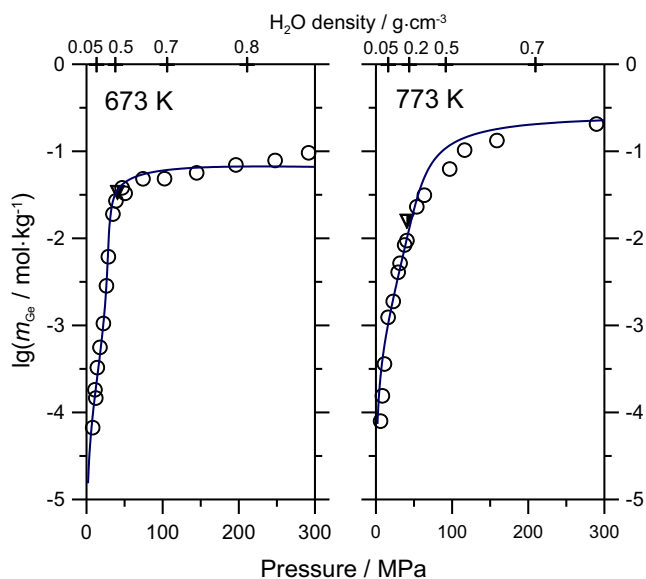


Fig. 6. Solubility of tetrahedral GeO_2 in pure water as a function of pressure at 673 and 773 K. Symbols correspond to the experimental data: $\circ$, Glemser et al. [20]; ∇ , Pokrovski et al. [8], lines are the model prediction.

of Glemser et al. [20] reported over a wide density range (0.02 – 0.80 g cm^{-3}). It can be seen that the HKF model predicts higher solubilities at 623 K and underestimates experimental solubility at 773 K while the density model systematically overestimates experimental solubility at all temperatures. In contrast, the new EoS matches the experimental data points within their accuracy over the whole T – P – ρ_1^* experimental range.

The equation-of-state parameters derived in this study allow prediction of the $\text{Ge}(\text{OH})_4$ partial thermodynamic properties as a function of T and P . For example, Fig. 8 shows the evolution of the partial molar volume $V_{\text{Ge}(\text{OH})_4(\text{aq})}$, and heat capacity $C_{p,\text{Ge}(\text{OH})_4(\text{aq})}$, as a function of temperature at 28 and 35 MPa. The decrease of both parameters to negative values while approaching the water critical point reflects the negative value of the Krichevskii parameter [18] and attests for the hydrophilic nature of $\text{Ge}(\text{OH})_4$, similar to that of other metalloid hydroxide complexes ($\text{Si}(\text{OH})_4$ [26]; $\text{B}(\text{OH})_3$ [36], $\text{As}(\text{OH})_3$ [37]), which exhibit so-called “electrolyte-type” behaviour [15,38].

The new thermodynamic data of $\text{Ge}(\text{OH})_4(\text{aq})$ together with those of $\text{Si}(\text{OH})_4(\text{aq})$ [15] allow quantitative comparisons between the two elements in high-temperature natural fluids. Despite strong similarities in the structure and energetics of aqueous germanic and silicic acids, differences in the thermodynamic properties of their oxide, germanate and silicate dissolution reactions may account for the increase of the Ge/Si ratio observed in high-temperature natural fluids, as was suggested by Pokrovski and Schott [6]. Fig. 9 depicts temperature dependencies of $\text{GeO}_2(\text{argutite})$ and $\text{SiO}_2(\text{quartz})$ solubilities at different pressures. Both minerals exhibit similar solubility patterns in water; however, the solubility of argutite shows a sharper temperature increase than that of quartz. This difference may be used in optimizing SiO_2 and GeO_2 crystal growth protocols in the lab [28,39,40], and may partly be responsible for the elevated Ge/Si ratios in high-temperature fluids emanating from the oceanic ridges [4,41,42].

Because the employed EoS is applicable both to dense liquid-like ($0.4 < \rho_1^* < 1.0 \text{ g cm}^{-3}$) and low-density vapour-like ($0.01 < \rho_1^* < 0.4 \text{ g cm}^{-3}$) aqueous fluids, it may also be used for predicting partitioning coefficients ($K_D(T, P_{\text{sat}})$, Eq. (4)) of Ge and Si between the coexisting vapour and liquid. Fig. 10 illustrates temperature dependencies of the computed K_D values for Ge and Si hydroxides.

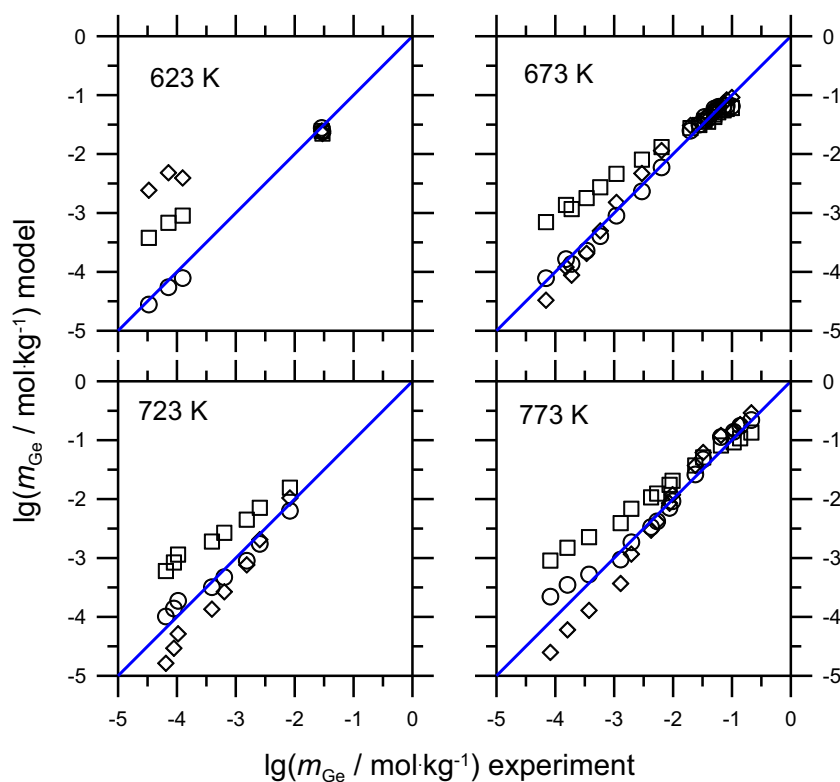


Fig. 7. Comparison of experimental [20] and model predictions ($\diamond$, HKF; $\square$, density model, both from [8]; $\circ$, the EoS model proposed in this study) for $\text{GeO}_2(\text{s})$ solubility for indicated temperatures. The range of water densities at each temperature is $0.02\text{--}0.8\text{ g cm}^{-3}$ and pressures $5\text{--}300\text{ MPa}$. The solid line corresponds to the ideal agreement and is drawn as an eye guide.

The predicted $\lg K_D(T, P_{\text{sat}})$ value at 623 K is -1.74 for $\text{Ge}(\text{OH})_4$, which is in agreement with the two estimations given in Section 3.2 at the same T - P point, $\lg K_D = -1.48 \pm 0.12$ and -1.82 ± 0.08 . It can also be seen that $\text{Ge}(\text{OH})_4$ is somewhat more hydrophilic (i.e., enriching the dense aqueous phase) compared to $\text{Si}(\text{OH})_4$.

This study demonstrates a high predictive potential of the proposed equation of state for different aqueous neutral species over a wide range of T - P -density parameters of natural and engineering fluids. Recently it has been successfully applied for description of hydroxides of metalloids ($\text{B}(\text{OH})_3$, $\text{As}(\text{OH})_3$, and $\text{Si}(\text{OH})_4$) [15]. Work is currently in progress to apply this EoS to other

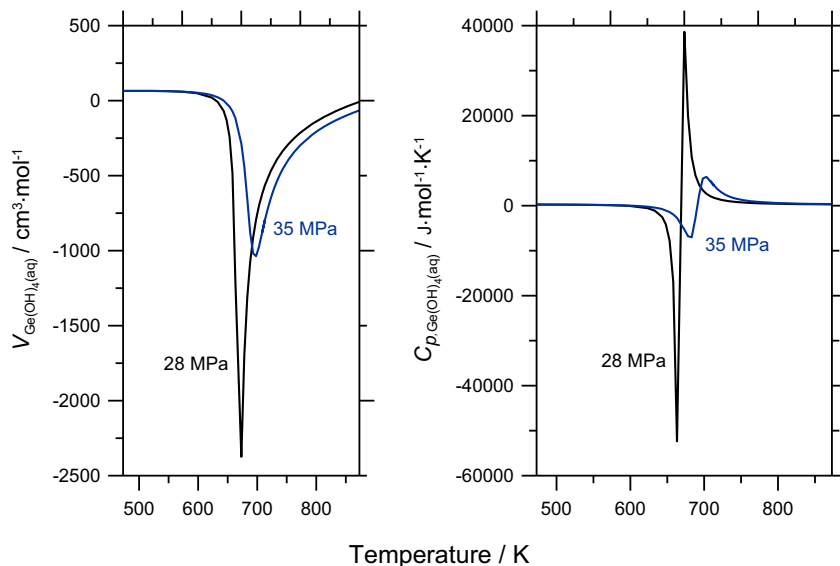


Fig. 8. Temperature dependence of the $\text{Ge}(\text{OH})_4(\text{aq})$ partial molar volume (left panel) and heat capacity (right panel) at 28 and 35 MPa . Curves are computed using the EoS parameters from Table 4.

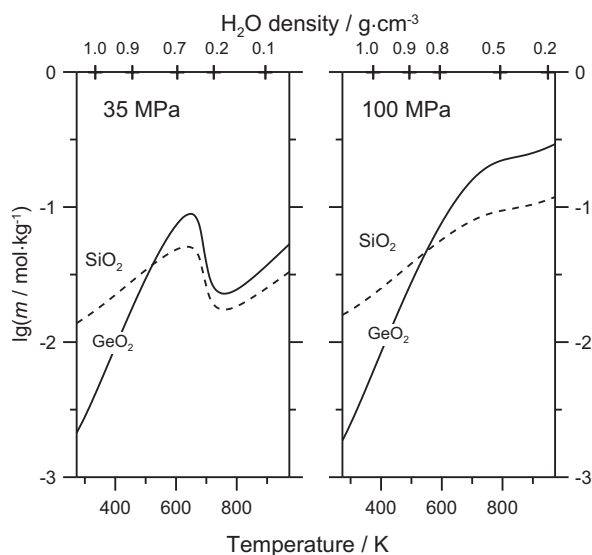


Fig. 9. Temperature dependence of GeO_2 (argutite) and SiO_2 (quartz) solubilities at 35 and 100 MPa. Curves were computed using the EoS parameters from Tables 3 and 4.

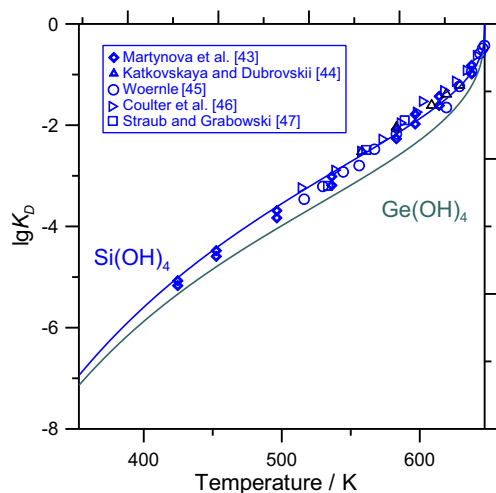


Fig. 10. Decimal logarithm of Si(OH)_4 and Ge(OH)_4 vapour–liquid distribution coefficient at infinite dilution, $\lg K_b$, at the water vapour saturation. Curves are computed using the EoS parameters from Table 4. Symbols represent the experimental data for Si(OH)_4 (♦, Martynova et al. [43]; ▲, Katkovskaya and Dubrovskii [44]; ○, Woernle [45]; □, Coulter et al. [46]; □, Straub and Grabowski [47]); no experimental data could be found for Ge(OH)_4 .

elements that form stable neutral species in aqueous solution (e.g., Sb, Ga, Se, Te).

Acknowledgments

This study was supported by the Russian Science Foundation (grant 14-17-00366 to NNA) and the Agence Nationale de la Recherche (grant SOUMET ANR-2011-Blanc SIMI 5-6 009 to GSP). We are grateful to the Associate Editor Dr. Ana Soto and two anonymous reviewers for their comments

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